Midterm Project: Income Classification Analysis

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Table of Contents

1. Introduction 3
2. Problem Definition and Model 3
   2.1 Task Definition 3
   2.2 Model Definition 4
3. Methodology 5
   3.1 Data Preparation 5
   3.2 Handling Outliers 5
   3.3 Variable Justification 6
   3.4 Data Splitting 7
   3.5 Resampling Method 8
   3.6 Evaluation Criteria 8
   3.7 Hypotheses 8
   3.8 Models and Results 9
   3.9 Discussion 10
4. Performance Metrics and Test Evaluation 11
   4.1 Performance Metrics (Validation Set) 11
   4.2 Performance on Test Dataset 12
5. Limitations and Future Work 14
6. Conclusion 15

1. Introduction

Understanding income levels is important for many real-world applications such as economic planning, taxation policy, and business decision-making. Organizations often need to classify individuals based on income to better understand workforce structure and financial behavior.

In this project, the objective is to classify whether an individual earns more than $50K per year using demographic and employment-related variables. The dataset contains information such as age, education, occupation, and working hours. These variables are expected to influence income.

To solve this problem, three classification models were used: logistic regression, Linear Discriminant Analysis (LDA), and decision tree. The models were trained and evaluated using a structured approach. Their performance was compared to identify the best model.

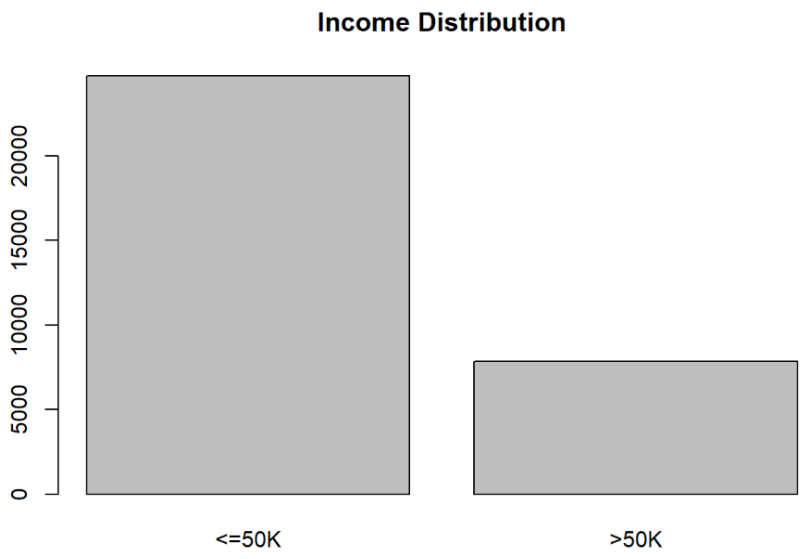
The results show that logistic regression performs better than the other models. It provides higher accuracy and more stable results across validation and test datasets. This report explains the full process, including data preparation, modeling, evaluation, and interpretation.

2. Problem Definition and Model

2.1 Task Definition

The goal of this project is to build a model that can predict whether an individual earns more than $50K per year. The input to the model consists of several variables such as age, education, occupation, and working hours. The output is a binary classification: either ≤50K or >50K.This problem is not simple because income depends on many factors that interact with each other. For example, education and occupation together influence income, but their effects are not always independent. In addition, the dataset is imbalanced (Plot 1), with more individuals in the lower income group. This makes it harder for models to correctly identify the minority class.

A good model should not only achieve high accuracy but also perform well on unseen data. It should be able to generalize beyond the training dataset.



Plot 1: Income Class Distribution

2.2 Model Definition

Three different models were used in this project.

Logistic regression is a commonly used model for binary classification. It estimates the probability that an observation belongs to a certain class. It works by modeling the relationship between the predictor variables and the log-odds of the outcome. This model is simple, interpretable, and effective when relationships are approximately linear.

Linear Discriminant Analysis (LDA) is another classification method. It finds a linear combination of variables that best separates the two classes. It assumes that the data follows a normal distribution and that both classes share a similar covariance structure.

The decision tree model works differently. It splits the data into smaller groups based on decision rules. Each split is based on a variable that best separates the classes. Decision trees are easy to understand but can be sensitive to noise and may overfit the data.

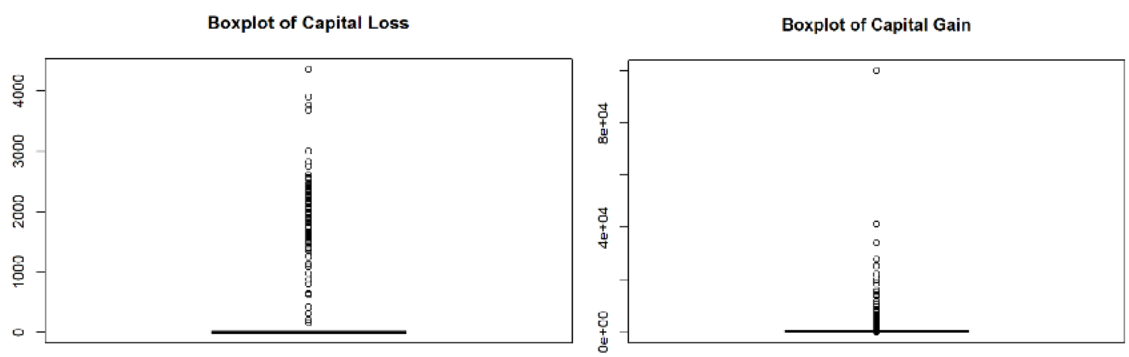
3. Methodology

3.1 Data Preparation

The dataset (Becker, B. & Kohavi, R.,1996) was first loaded into R, and all variables were initially treated as character type. This approach was used to prevent incorrect automatic type conversion and to allow full control over preprocessing steps. Missing values in the dataset were represented by the symbol "?". These values were identified and converted into NA so that they could be handled properly. Rows containing missing values were removed using listwise deletion. This method was chosen because the dataset remained sufficiently large after cleaning, and removing incomplete observations helped maintain consistency in the analysis. After handling missing values, variables were converted into their appropriate data types. Numeric variables such as age, education_num, capital_gain, capital_loss, and hours_per_week were converted into numeric format. Categorical variables such as workclass, education, occupation, and income were converted into factor type. This ensured that the models interpret the variables correctly.

3.2 Handling Outliers

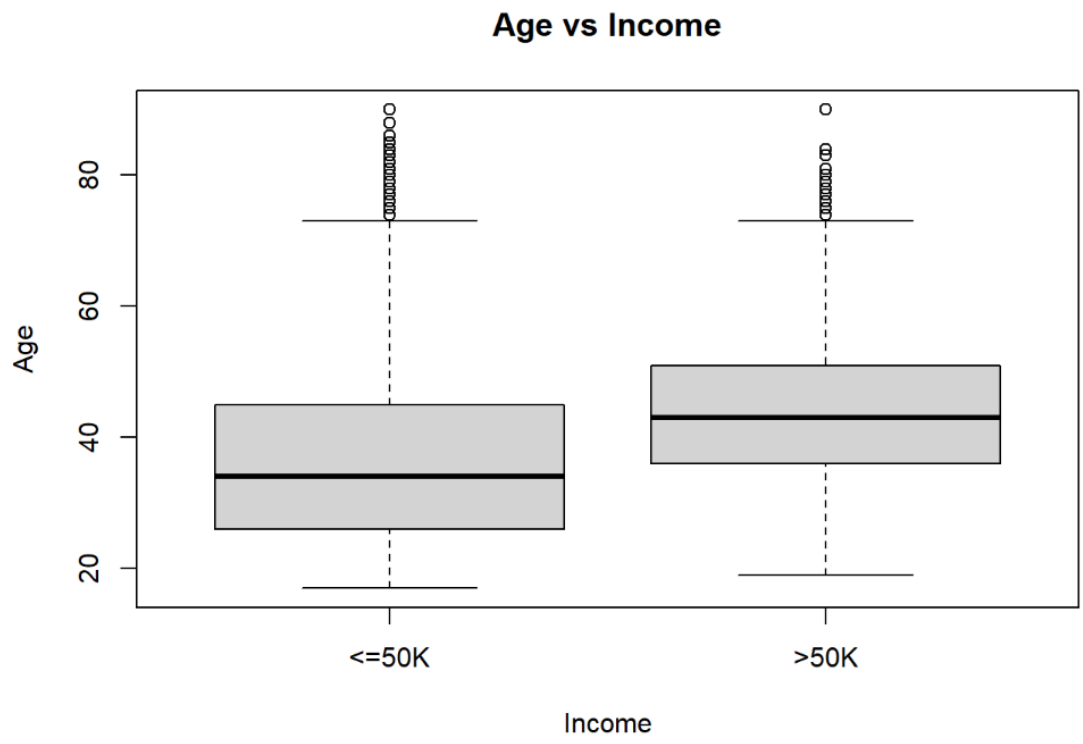
Outliers were examined using boxplots for numeric variables such as age, capital_gain, capital_loss, and hours_per_week. The analysis showed that variables such as capital_gain contains extreme values whereas capital _loss contains strong outliers (Plot 2). Most observations are concentrated near zero, while a small number of observations have very large values.



Plot 2: Visualisation of outliers by Boxplot
(The boxplot clearly shows the presence of extreme values in capital_gain. Most observations are concentrated near zero, while a small number of observations have very high values. These extreme values represent real financial situations rather than data errors. Therefore, they were retained in the dataset to preserve

3.3 Variable Justification

The variables used in this study were selected based on their relevance to income determination. The variable fnlwgt was excluded from the model because it represents a sampling weight rather than a meaningful predictor of income. It reflects the relative importance of observations in the survey design and does not describe individual characteristics. The boxplot indicates that individuals in the >50K category tend to be older on average than those in the ≤50K category. This suggests that age, which may reflect work experience, has an important relationship with income classification (see Plot 3). Education and education_num represent skill level and qualification, which are strong indicators of earning potential.

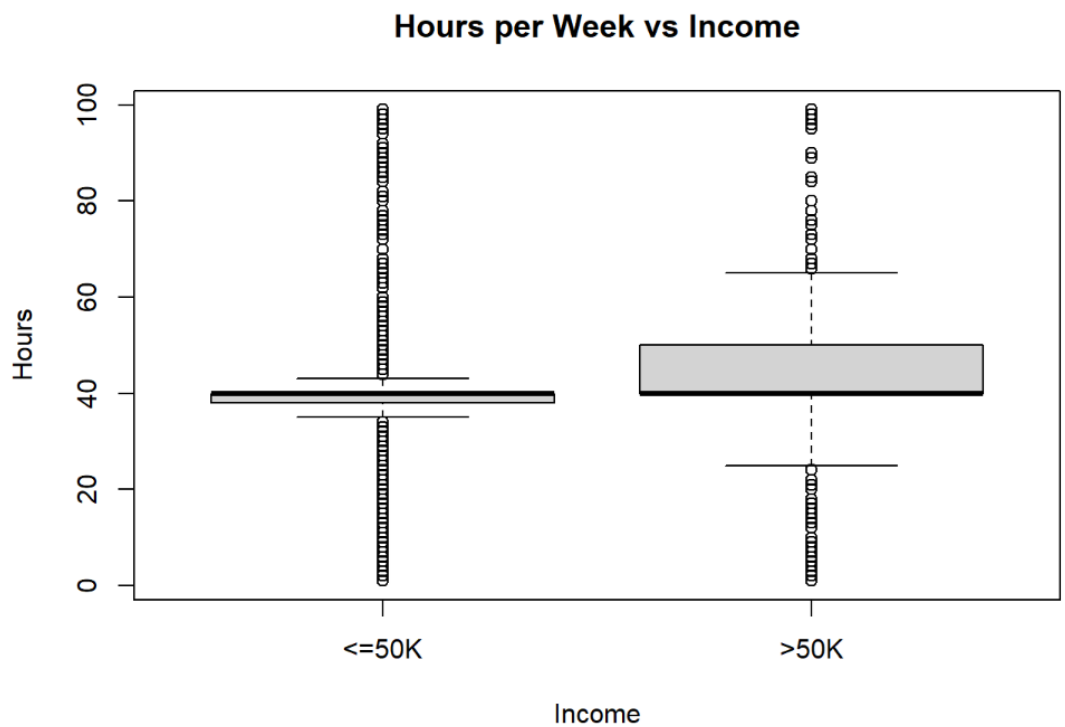


Plot 3: Income vs Age

Occupation and workclass describe the type of job and employment structure. Professional and managerial roles are generally associated with higher income compared to service or manual jobs. Relationship status may also influence income, as married individuals often have more stable economic conditions.

Hours_per_week represents the amount of work performed and is directly related to income. The boxplot shows that individuals earning more than $50K generally work more hours per week than

those in the lower income category. This suggests that working hours are positively associated with higher income (see Plot 4). Capital_gain and capital_loss provide information about financial activity, which may indicate higher income levels.



Plot 4: Hours per week vs income

These variables were chosen before modeling based on their logical and practical relevance. This helps avoid selecting variables only based on statistical significance and ensures that the model remains meaningful and interpretable.

3.4 Data Splitting

The cleaned dataset was divided into two subsets: a training set and a validation set. A 70/30 split was used, where 70% of the data was used for training and 30% was used for validation.

This split was chosen to balance model learning and performance evaluation. A larger training set allows the model to learn patterns more effectively, while a sufficiently large validation set provides a reliable estimate of model performance.

In addition to this split, a separate dataset (adult.test) was used as a held-out test set. This dataset was not used during model training or validation, which ensures that it provides an unbiased evaluation of the final model.

3.5 Resampling Method

To improve the robustness of the models, bootstrap resampling with 25 repetitions was applied during training using the caret framework.

Bootstrap resampling involves repeatedly sampling observations from the training dataset with replacement. Each sample is used to train the model, and the results are averaged across all repetitions. This reduces the dependence on a single training dataset and provides a more stable estimate of model performance. This method helps ensure that the model is not overly sensitive to specific data splits and improves the reliability of the results.

3.6 Evaluation Criteria

The performance of the classification models was evaluated using multiple metrics, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of predictions. Precision measures how many observations predicted as the positive class (>50K) are actually correct. Recall (sensitivity) measures how well the model identifies the positive class. The F1-score provides a balance between precision and recall (James et al., 2021).

Because the dataset is moderately imbalanced, with a larger proportion of observations in the ≤50K category, accuracy alone may not provide a complete evaluation of model performance. A model can achieve high accuracy by correctly predicting the majority class while performing poorly on the minority class. Therefore, model selection in this study is not based solely on accuracy. Precision, recall, and F1-score are also considered to ensure a more balanced and reliable evaluation of model performance.

3.7 Hypotheses

It is expected that logistic regression will perform better than the other models due to its suitability for binary classification problems and its ability to handle both numeric and categorical variables effectively.

LDA is expected to perform reasonably well but may be affected by violations of normality assumptions and skewed variables.

The decision tree model is expected to have lower performance due to its sensitivity to class imbalance and tendency to favor the majority class.

Overall, it is hypothesized that logistic regression will provide the most balanced performance across evaluation metrics.

3.8 Models and Results

Three classification models—logistic regression, Linear Discriminant Analysis (LDA), and decision tree were trained and evaluated using the validation dataset. Model performance was compared using multiple metrics, including accuracy, precision, recall, and F1-score. (see Table 1).

Model	Accuracy	Precision	Recall	F1_Score
Logistic Regression	0.8494	0.8778	0.9288	0.9026
LDA	0.8363	0.8665	0.9245	0.8946
Decision Tree	0.8030	0.7947	0.9946	0.8835

(Table 1: Model Comparison on Validation Data with Evaluation Metrics)

The results show that logistic regression achieves the best overall performance across all evaluation metrics. This is consistent with findings in statistical learning literature where logistic regression performs well in binary classification problems (James et al., 2021). It provides a strong balance between precision and recall, resulting in the highest F1-score among the three models. This indicates that logistic regression is effective in correctly identifying both income classes.

The LDA model performs reasonably well but shows slightly lower performance across these metrics. This may be due to its sensitivity to distributional assumptions and the presence of skewed variables such as capital_gain and capital_loss.

The decision tree model shows the lowest performance, particularly due to its tendency to favor the majority class (≤50K). This results in weaker precision and recall for the minority class (>50K).

Overall, logistic regression was selected as the final model because it provides the most balanced and reliable performance across all evaluation metrics, rather than relying solely on accuracy.

3.9 Discussion

The comparison based on accuracy, precision, recall, and F1-score shows that logistic regression provides the most balanced performance among the three models. The results of this study show clear differences in the performance of the three models. Logistic regression achieved the highest validation accuracy, followed by LDA, while the decision tree model showed the lowest performance. These results can be explained by considering both the characteristics of the data and the assumptions of each model.

Logistic regression performed best because it is well suited for binary classification problems and can handle both numeric and categorical variables effectively. The relationship between several predictors, such as education, occupation, and hours worked, appears to be approximately linear when transformed into log-odds. This makes logistic regression a good fit for the data. In addition, logistic regression is relatively robust to issues such as multicollinearity and does not rely on strict distributional assumptions, which helps improve its performance.

The LDA model also performed reasonably well, but slightly worse than logistic regression. One possible reason is that LDA assumes that the predictor variables follow a normal distribution and that both classes share the same covariance structure. In this dataset, some variables, such as capital_gain and capital_loss, are highly skewed and contain extreme values. These characteristics violate the assumptions of LDA, which reduces its effectiveness. Additionally, the presence of categorical variables with many levels may introduce instability in the estimation of class boundaries.

The decision tree model showed the lowest accuracy among the three models. This can be explained by its sensitivity to class imbalance and its tendency to favor the majority class. In this dataset, most observations belong to the ≤50K category. As a result, the decision tree tends to create splits that improve accuracy for the majority class while ignoring the minority class. This behavior is reflected in the model's high sensitivity but low specificity. While the decision tree is capable of capturing nonlinear relationships, its performance is limited when the data is imbalanced and contains many categorical variables.

Another important observation is the close similarity between validation and test performance across evaluation metrics. The results remain consistent when comparing accuracy, precision, recall, and F1-score. This indicates that the model generalizes well to unseen data and does not suffer from significant overfitting. This indicates that the model generalizes well to unseen data and does not suffer from significant overfitting. The use of a 70/30 data split and bootstrap resampling contributed to this stability by providing both sufficient training data and reliable evaluation.

The effect of outliers also plays a role in model performance. Variables such as capital_gain contain extreme values, but these values were retained because they represent real financial situations. Logistic regression is able to handle these values more effectively than LDA, which is more sensitive to distributional assumptions. This further explains why logistic regression outperformed LDA.

Overall, the results show that model performance depends not only on the algorithm but also on the nature of the data. Logistic regression performs best because it balances flexibility, robustness, and interpretability. The comparison across models highlights the importance of selecting an appropriate method based on data characteristics rather than relying on a single approach.

4. Performance Metrics and Test Evaluation

4.1 Performance Metrics (Validation Set)

The performance of the selected model, logistic regression, was evaluated using the validation dataset. Multiple evaluation metrics were used, including accuracy, precision, recall, and F1-score, to provide a comprehensive assessment of model performance (see Table 2).

Metric	Value
Accuracy	0.8494
Precision	0.8778
Recall	0.9288
F1-score	0.9026

(Table 2: Table 2: Performance Metrics of Logistic Regression on Validation Data)

The model shows strong performance across all evaluation metrics. The accuracy indicates high overall correctness of predictions. Precision reflects the model's ability to correctly identify individuals in the higher income category (>50K), while recall indicates how effectively the model captures this minority class. The F1-score confirms that the model maintains a good balance between precision and recall.

Using multiple evaluation metrics ensures that the model is not only accurate but also reliable in identifying both classes, particularly in the presence of class imbalance.

4.2 Performance on Test Dataset

To evaluate the generalization ability of the selected model, logistic regression was applied to a separate test dataset that was not used during model training or validation. This provides an unbiased assessment of model performance on unseen data.

The performance of the model on the test dataset is summarized in Table 3. The model maintains strong predictive performance, indicating that it generalizes well beyond the training data. The evaluation metrics show that the model performs effectively in classifying the majority class (≤50K), while performance for the minority class (>50K) is relatively lower due to class imbalance.

Metric	Value
Accuracy	0.8467
Precision	0.8767
Recall	0.9271
F1-score	0.9012

(Table 3: Performance Metrics of Logistic Regression on test data)

The results also show that though the evaluation metrics slightly degrade on test data, they remain relatively consistent across the two datasets (see Table 4). This consistency indicates that the model does not suffer from significant overfitting and performs reliably on unseen data.

Metric	Logistic Model's Validation Performance	Logistic Model's Test Performance

Accuracy	0.8494	0.8467
Precision	0.8778	0.8767
Recall	0.9288	0.9271
F1-score	0.9026	0.9012

(Table 4: Performance comparison across validation and test data)

The confusion matrix provides a detailed and comprehensive evaluation of the model's classification performance beyond overall accuracy. It reveals how predictions are distributed across the two income classes and highlights the types of errors made by the model. The results show that the model correctly classifies a large number of observations in the ≤50K category, indicating strong performance for the majority class. This is reflected in high sensitivity, which demonstrates the model's effectiveness in identifying individuals with lower income. However, the model exhibits comparatively weaker performance in correctly identifying the >50K category, as indicated by lower specificity.

		Actual	
		<= 50K	>50K
Prediction	<=50K	10532	1481
	>50K	828	2219

(Table 5: Confusion Matrix of Logistic Model on test data)

This suggests that a number of high-income individuals are misclassified as belonging to the lower income group, resulting in false negatives. This pattern is largely influenced by the class imbalance present in the dataset, where the majority of observations belong to the ≤50K category. As a result, the model tends to favor the majority class, leading to better performance for that class at the expense of the minority class. Despite this limitation, the confusion matrix confirms that the model maintains strong overall predictive capability while also highlighting areas where performance can

be improved, particularly in detecting the minority class. The use of confusion matrices and derived metrics provides a more comprehensive evaluation of classification models beyond accuracy (James et al., 2021).

Overall, the results confirm that the logistic regression model achieves stable and reliable performance across both validation and test datasets, making it suitable for income classification tasks.

5. Limitations and Future Work

One limitation of this study is the class imbalance present in the dataset, where the majority of observations belong to the ≤50K category. This imbalance affects the model's ability to correctly identify the minority class (>50K), as reflected in the confusion matrix and lower specificity. As a result, the model tends to favor the majority class, which can reduce its effectiveness in identifying higher-income individuals.

Another limitation is the use of relatively simple classification models. While logistic regression performed well, more advanced models such as random forests, gradient boosting, or ensemble methods could potentially improve performance, especially in handling complex relationships and class imbalance.

In addition, some predictor variables, such as capital_gain and capital_loss, exhibit strong skewness and contain extreme values. Although these values represent real financial conditions and were retained in the analysis, further transformation or feature engineering could improve model performance.

Future work could focus on addressing class imbalance using techniques such as resampling (oversampling or undersampling), class weighting, or synthetic data generation methods like SMOTE. Additionally, exploring advanced models and incorporating interaction effects or feature transformations may further enhance predictive accuracy and overall model performance.

6. Conclusion

This study developed classification models to predict whether an individual earns more than $50K per year using demographic and employment-related variables. Three models—logistic regression, Linear Discriminant Analysis (LDA), and decision tree—were implemented and evaluated using a structured methodology.

The models were compared using multiple evaluation metrics, including accuracy, precision, recall, and F1-score, to ensure a comprehensive assessment of performance. The results show that logistic regression provides the best overall performance across these metrics, demonstrating a strong balance between precision and recall.

The model also maintained consistent performance between validation and test datasets, indicating good generalization and minimal overfitting. The confusion matrix analysis further revealed that while the model performs well for the majority class (≤50K), its performance for the minority class (>50K) is comparatively lower due to class imbalance.

Overall, the findings highlight the importance of using multiple evaluation metrics rather than relying solely on accuracy, especially in imbalanced classification problems. Logistic regression was identified as the most suitable model for this task, offering a combination of predictive performance, stability, and interpretability.

In conclusion, this study demonstrates that a well-structured approach to data preprocessing, model evaluation, and interpretation can lead to reliable and meaningful classification results, while also providing insights into areas for future improvement.

REFERENCES

Becker, B. & Kohavi, R. (1996). Adult [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5XW20.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: with Applications in R* (2nd ed.). Springer.